Philosophy of Space and Time

Lecture 13: *General Relativity I — The Equivalence Principle*

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Today's Plan

Where We Left Off

Last lecture closed with a question:

- Minkowski spacetime drops absolute simultaneity from Galilean spacetime
- But it still has a **flat metric, fixed once and for all**, independent of what matter does
- Could this fixed background also be excessive structure?

Today we begin the answer. Start with an empirical fact: the **equality of inertial and gravitational mass**.

Today's Argument in One Slide

1. In Newtonian physics, $m_I = m_G$ is a brute coincidence
2. Einstein: theory should explain this
3. Locally, accelerated frames in flat spacetime are indistinguishable from inertial frames in a gravitational field — the **equivalence principle**
4. Globally, **tidal forces** reveal genuine **spacetime curvature**
5. Free-falling bodies trace **geodesics** of this curved geometry — and so the metric itself must respond to matter

This last step — a *dynamical* metric — is the crucial new feature of GR

An Empirical Coincidence

Two Notions of Mass

Newtonian physics distinguishes two roles that "mass" plays:

Inertial mass m_I

The quantity that appears in $F = m_I a$. Measures the resistance of a body to being accelerated (push a body, see how much it accelerates).

Gravitational mass m_G

The quantity that appears in $F_{\text{grav}} = G\, m_G\, M_G / r^2$. Measures how strongly a body produces and responds to gravity.

Conceptually distinct, like "electric charge" and "mass".

Galileo's Observation

In a uniform gravitational field, all bodies fall at the same rate.

- Acceleration of a falling body: $a = F/m_I = (m_G/m_I)\,g$
- If a is the same for every body, then m_G/m_I is the same for every body

This is the **universality of free fall** — one of the oldest and best-tested facts in physics.

Newton's Pendulum Experiment

Newton himself was aware of how striking the equality was. He confirmed it to part-in-10^3 with pendulums of different materials swinging in matched bobs.

> *By experiments made with the greatest accuracy, I have always found the quantity of matter in bodies to be proportional to their weight…*
>
> — Newton, *Principia*, Book III, Prop. 6

Newton accepted the equality as fact. In his framework it had no deeper explanation.

From Eötvös to Modern Tests

Modern tests have pushed the precision much further.

- Eötvös (1890s): $|m_G/m_I - 1| < 10^{-9}$
- Lunar laser ranging: Earth and Moon fall toward the Sun with equal accelerations to $\sim 10^{-13}$
- MICROSCOPE satellite (2017–): $\sim 10^{-15}$

No deviation has ever been found. Whatever explains $m_I = m_G$ does so to extraordinary precision, across every kind of matter ever tested.

Why This Is Striking

In Newtonian physics, the equality has no structural rationale.

- Electric charge is not equal to inertial mass — different bodies have different charge-to-mass ratios
- Why should gravitational “charge” alone be locked to inertial mass?
- As far as Newtonian theory is concerned, gravity could perfectly well act on different bodies with different strengths

?? *A robust empirical regularity that the theory does not explain is a clue that the theory misses something important. What is the equality of m_I and m_G trying to tell us?*

Einstein's Happiest Thought

Bern, 1907

Einstein later recalled the moment he saw the way forward.

> *I was sitting in a chair in the patent office at Bern when all of a sudden a thought occurred to me: "If a person falls freely, he will not feel his own weight." I was startled. This simple thought made a deep impression on me. It impelled me toward a theory of gravitation.*
>
> — Einstein, recalling 1907

The free-faller feels no gravity — because in a sense, there is no gravity to feel.

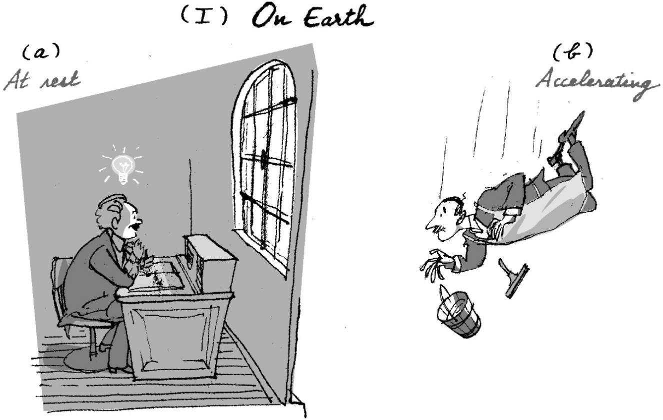
Why This Matters

Galileo's universality of free fall has a striking consequence:

- In a freely falling laboratory, all bodies fall together
- Releasing a ball, it floats next to you — it does not "fall"
- Mechanical experiments inside the laboratory yield the same results as if there were no gravitational field at all

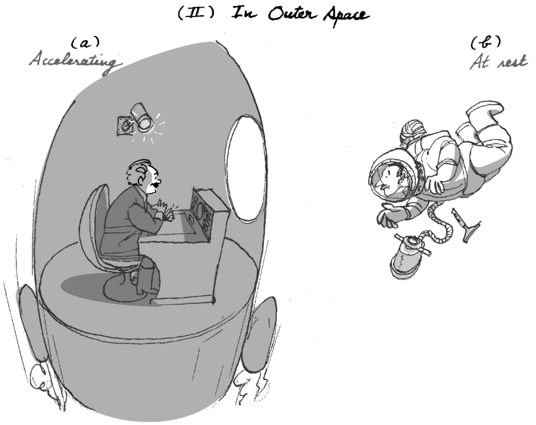
Locally, free fall in gravity is indistinguishable from inertial motion in flat space.

Analog of Galileo's Ship?



(Images from Janssen (2014), in *Cambridge Companion to Einstein*)

And the Other Direction



(Images from Janssen (2014), in *Cambridge Companion to Einstein*)

Implications

?? *Either observer can claim to be at rest, they then disagree about whether there is gravity present. (I) and (II) can be accounted for either with gravity or with acceleration. Contrast between freely-falling and non-freely falling motion, not inertial vs. accelarted motion.*

The Equivalence Principle

Equivalence Principle

The trajectory of a freely falling *test particle* (small mass, no charge) depends only on its initial position and velocity — not on its internal composition.

- This is just the universality of free fall, restated
- Implies $m_I = m_G$ (up to a universal constant)
- In itself, compatible with Newtonian gravity

Einstein's Version

Einstein's stronger conjecture:

Strong Equivalence Principle

In any sufficiently small region of spacetime, the laws of physics — *all* of them, not just mechanics — take the form they have in special relativity, in some appropriately chosen frame.

- Extends the equivalence to all of physics, including electromagnetism
- "Sufficiently small" is doing important work — we will see why

What the Elevator Shows

Einstein's elevator thought experiment motivates the principle.

- Sealed inside an elevator with no windows, released objects fall to the floor at constant acceleration
- Cannot tell whether the elevator is sitting in a gravitational field, or being towed through gravity-free space
- Whatever experiment performed, the two scenarios give the same answer

Whether "accelerating" or "in a gravitational field" is, locally, a distinction without a difference.

What the EP Does and Does Not Show

The EP shows

Locally, gravity and acceleration are interchangeable. No *local* experiment can distinguish them.

The EP does not show

That there is no real gravity in the world, or that gravity is a mere coordinate effect.

In fact, gravity *cannot* be uniformly transformed away over an extended region: real gravitational fields produce **tidal effects** that no choice of frame can eliminate.

Tidal Forces and Curvature

Where the Equivalence Breaks Down

Two test particles, released side by side in a gravitational field, do not fall along parallel trajectories.

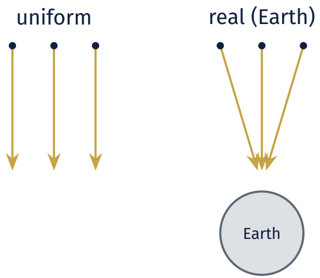
- Gravitational fields point toward the source mass (e.g. the center of the Earth)
- Two objects falling toward Earth converge toward each other as they fall
- Two objects spaced vertically experience different field strengths — the lower one accelerates faster, and the spacing increases

These are **tidal forces**.

Tidal Forces in Pictures

- In a uniform field (or accelerated frame), released particles fall on parallel paths
- In a real gravitational field, released particles converge or diverge depending on orientation

  (Called tidal forces because this is what raises produces ocean tides.)



Geodesic Deviation

In SR / Minkowski spacetime, two initially parallel inertial worldlines remain parallel forever.

- This is a property of the flat metric
- Parallel free-fall trajectories should remain parallel

But in a real gravitational field, initially parallel free-fall trajectories converge or diverge.

The conclusion Einstein drew

The geometry must be *not flat*. The convergence of free-fall trajectories is a hallmark of **curvature** of spacetime itself, just as initially parallel great circles on a sphere converge.

Curvature as the Real Signature

The equivalence principle and the existence of tidal forces sit together coherently:

- **Locally** (in any small enough region), gravity and acceleration are indistinguishable — the geometry looks flat
- **Globally** (over extended regions), tidal forces reveal that the geometry is in fact curved
- The flat metric of SR is a *local* approximation; the full geometry of a gravitating spacetime is curved

"Real gravity" = curvature that no choice of frame can transform away.

Free Fall as Inertial Motion

The Reinterpretation

In Newtonian physics:

- Inertial motion = motion in a straight line at constant speed (no forces)
- A falling apple is acted on by the force of gravity
- "Free fall" is a *forced* trajectory

Einstein's reinterpretation:

- Free fall is the new **inertial** motion
- There is no force of gravity
- Apples and planets are just following the closest thing to a "straight line" in curved spacetime — a **geodesic**

Geodesics, Intuitively

In flat geometry, a straight line is the curve of shortest length between two points.

- On a sphere, the analog is a great circle
- On any curved surface, geodesics are the locally straightest curves
- In Minkowski spacetime, timelike geodesics maximize proper time (recall the twins)
- In a curved spacetime, timelike geodesics generalize this — they are the worldlines of free-fall

Why Gravity Is Not a Force

The geometric reinterpretation

A body in free fall is not being pushed or pulled by anything. It follows a geodesic of the (curved) spacetime around it — the relativistic analog of inertial motion. A *force* is precisely what would be needed to make a body deviate from a geodesic.

- Standing on the ground, you feel weight because the floor is pushing you off the geodesic you would otherwise follow
- The Moon orbits the Earth not because gravity "pulls" it but because it follows the closest thing to a straight line in the curved spacetime around Earth

What This Costs Us

The reconception is not free. Some intuitions have to go.

- “Force” was the central category of Newtonian mechanics
- The geometry of spacetime now does the work that “the force of gravity” used to do
- And the geometry has to vary, *respond* to matter — so that it can vary in different regions

The Metric
Becomes Dynamical

The Novel Idea

In Minkowski spacetime, the metric is a fixed background — given once and for all, the same in every case.

- In GR, the metric is a **dynamical field** that varies from place to place
- The amount of matter determines the amount of curvature
- The geometry of spacetime is not given in advance — it is part of what physics has to determine

Einstein's field equations (preview)

A precise relation between matter (and energy) and curvature, which we will look at next time. For today: the metric is no longer a backdrop; it is one of the variables.

The Elimination Story Continued

Where this fits in the overall arc:

1. Newton $\rightarrow$ Galilean: drop persistence of spatial points
2. Galilean $\rightarrow$ Minkowski: drop absolute simultaneity
3. Minkowski $\rightarrow$ **Einsteinian**: drop the *fixed* metric

Each step removes a structure that the new physics shows to be excessive. GR's contribution is the most radical: it removes the fixed background itself.

What's New, What's Preserved

Preserved from SR:

- Local light-cone structure
- Local Lorentz invariance
- Proper time along worldlines
- Free fall as inertial motion (now generalized)

New in GR:

- The metric is a field, not a background
- Spacetime is curved by matter
- No global inertial frames
- Gravity is geometry, not a force

Summary

1. $m_I = m_G$ is a precise empirical fact left unexplained in Newtonian physics
2. **Einstein's happiest thought**: free fall feels weightless — locally, gravity and acceleration are indistinguishable
3. **Equivalence principle**: universality of free fall, and local validity of special relativity
4. **Tidal forces** reveal that gravity cannot be globally transformed away — the signature of **spacetime curvature**
5. **Free fall as inertial motion**: geodesics replace "the force of gravity"
6. **The metric becomes dynamical**

For Next Time

- **Read:** Maudlin, Ch. 6 (continued); Geroch, Chs. 7–8.
- **Think about:**
  - How could one give a precise statement of “the metric responds to matter”?
  - In what sense is GR’s metric “less” structure than Minkowski’s flat one?
  - In what sense might it be *more*?

May 19: **General Relativity II — Curved Spacetime and the Field Equations**

- Geodesics in curved spacetime
- Curvature, intuitively and formally
- Einstein’s field equations: matter $\leftrightarrow$ geometry
- Background independence and what it does (and does not) settle

Overflow: The Pound–Rebka Experiment

Robert Pound and Glen Rebka, Harvard, 1959–1960: measured **gravitational redshift** in a tower at Jefferson Lab.

- Photons emitted at the bottom of the tower, absorbed at the top, lose energy — their frequency drops
- The shift is $\sim 2.5 \times 10^{-15}$ per meter of altitude
- Pound and Rebka measured it using the Mössbauer effect with ^{57}Fe gamma rays
- The first laboratory confirmation of an EP-derived prediction within a single building

Einstein had predicted the effect in 1907 from the equivalence principle alone.

Questions?

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